Imperial College London

Lecture 5

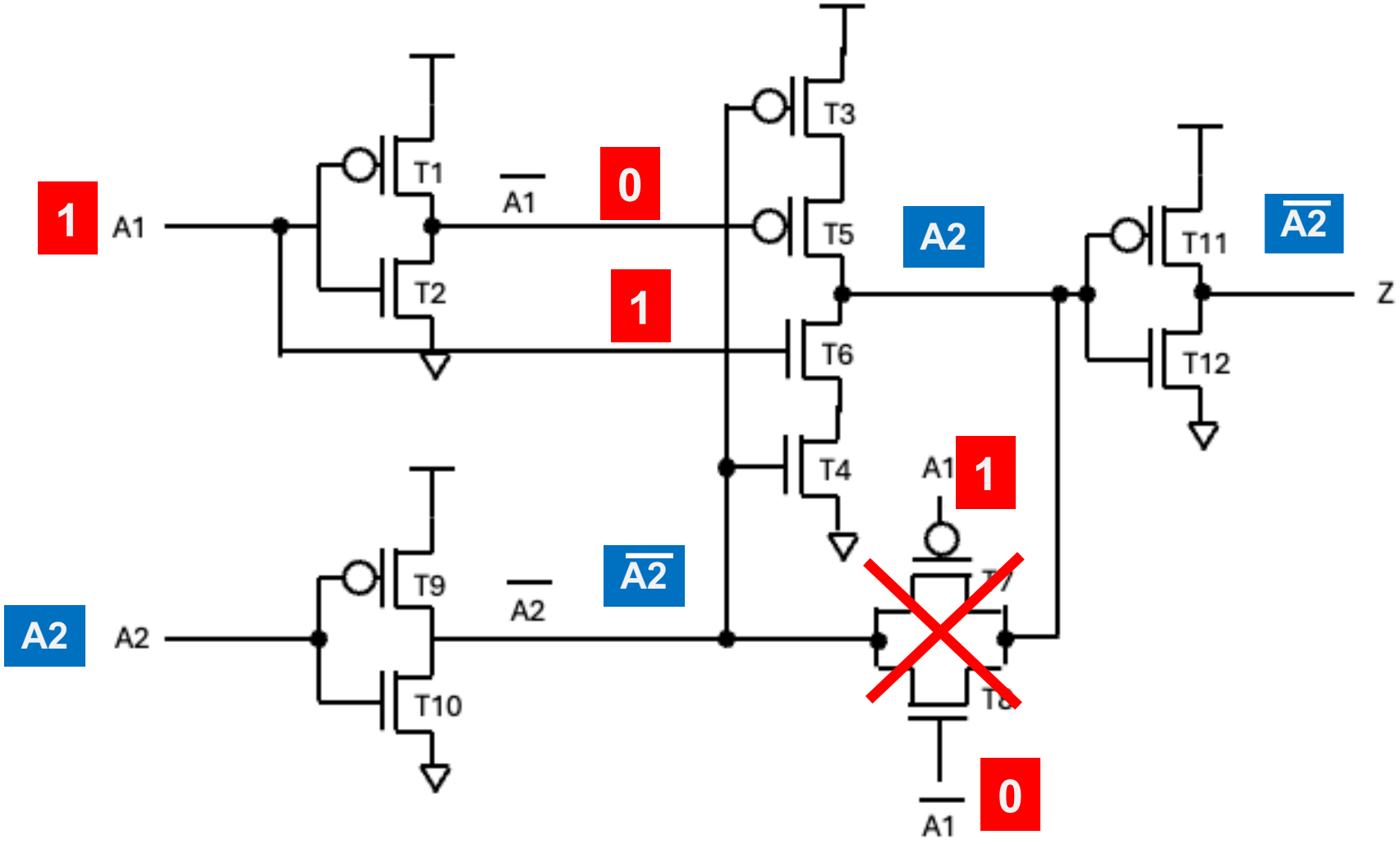
Combinational Circuits

Prof Peter YK Cheung

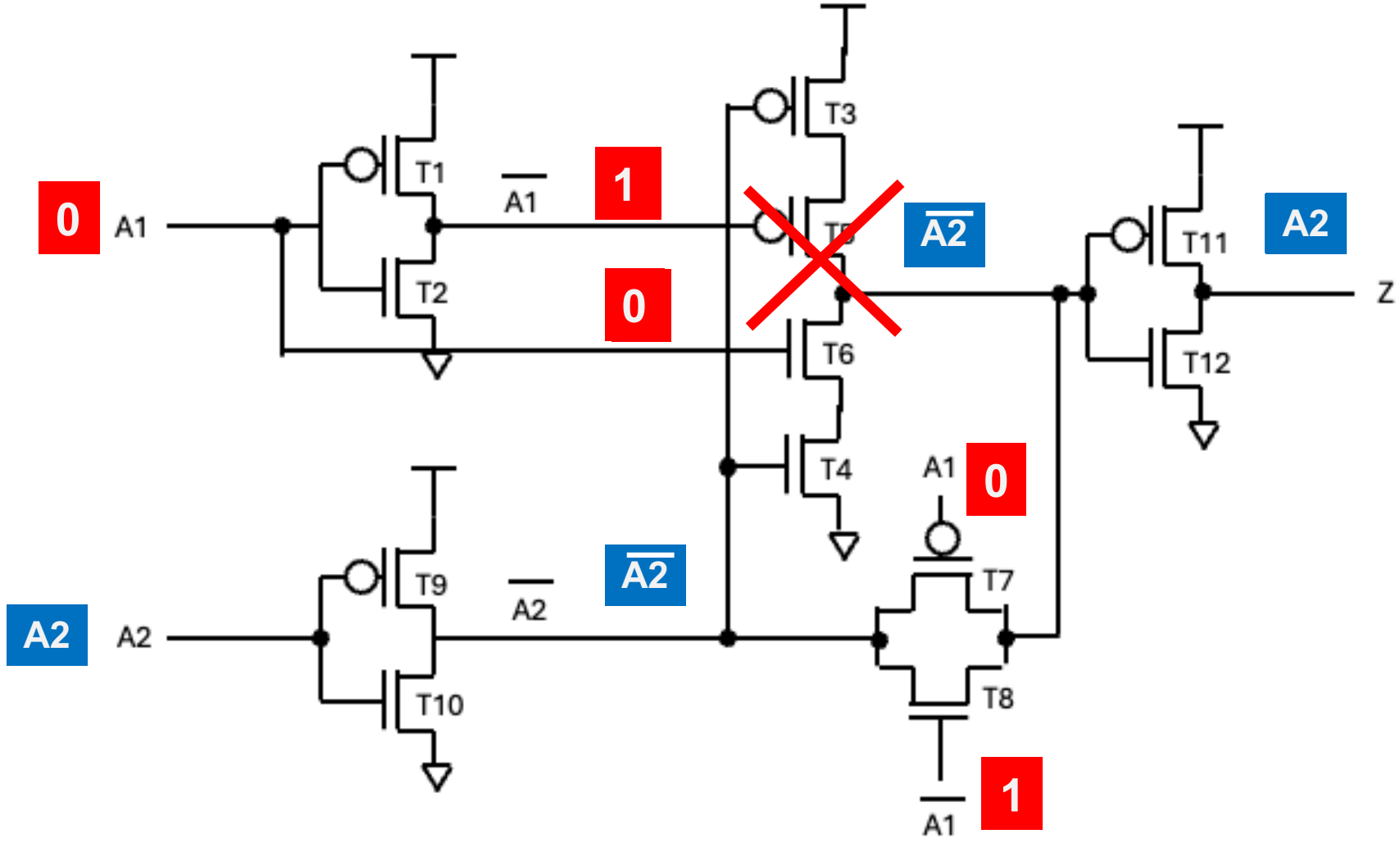
Department of Electrical & Electronic Engineering

URL: www.ee.ic.ac.uk/pcheung/teaching/EE4-VLSI/
E-mail: p.cheung@imperial.ac.uk

XOR gate schematic



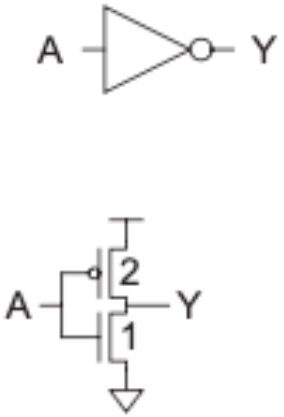
XOR gate schematic



Compound Gates

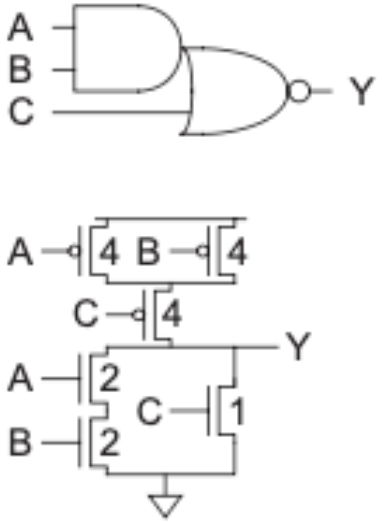
Unit Inverter

$Y = \overline{A}$



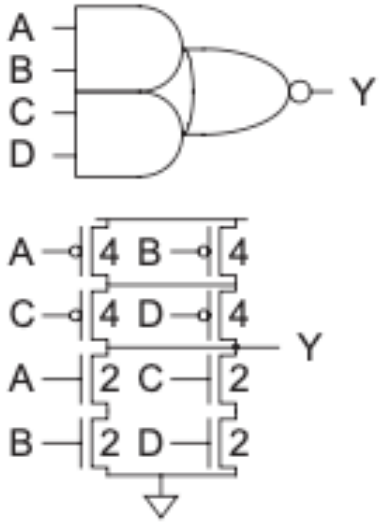
AOI21

$Y = \overline{A \cdot B + C}$



AOI22

$Y = \overline{A \cdot B + C \cdot D}$



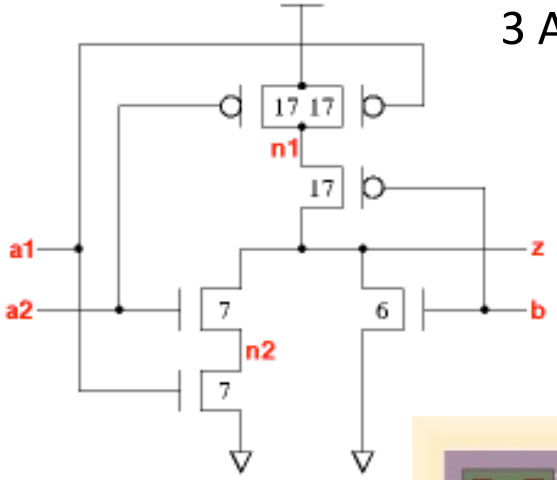
W&H p330

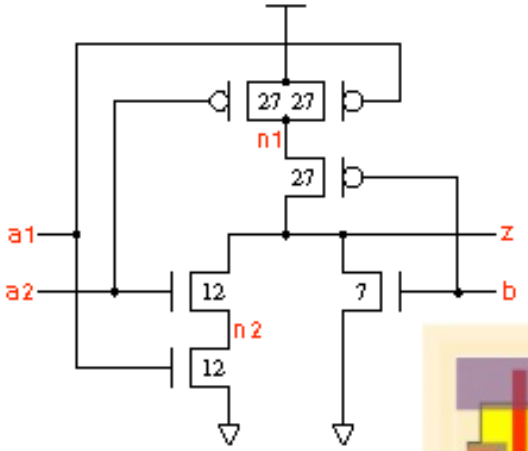
vsclib Standard Cell Library – aoi21

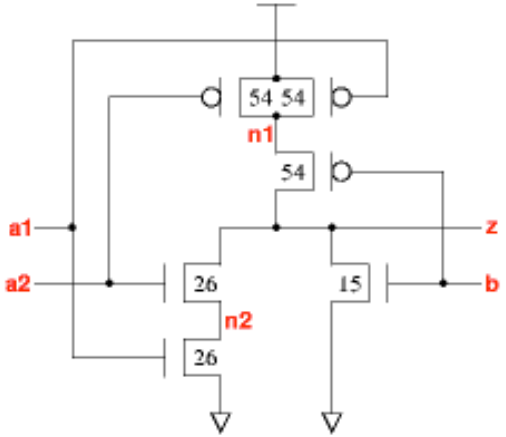


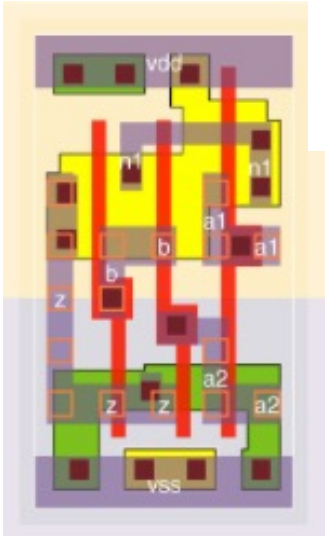

```
z:
((a1*a2)+b)'
```

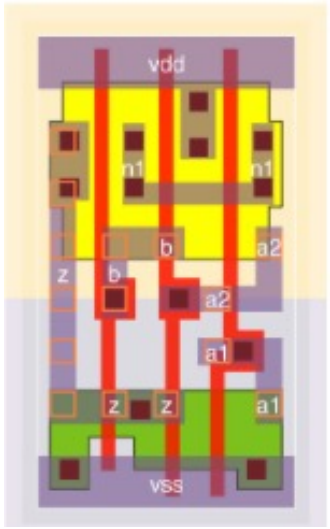
3 AOI31 cells with different drive strengths.

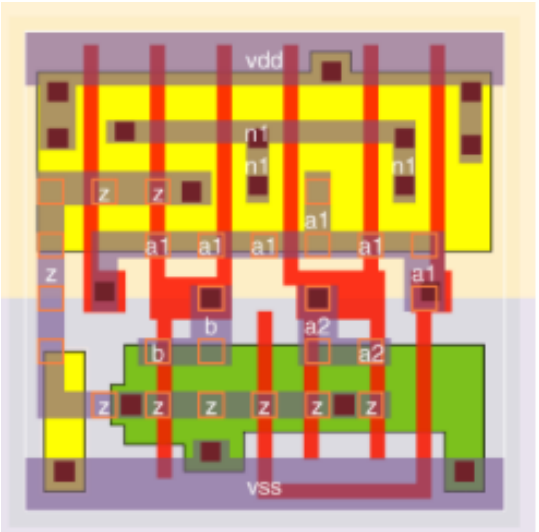




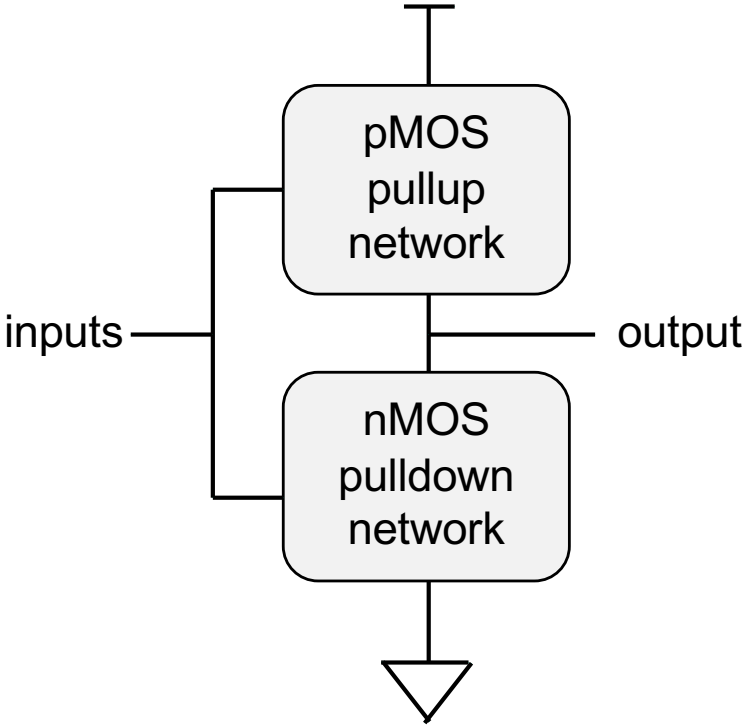


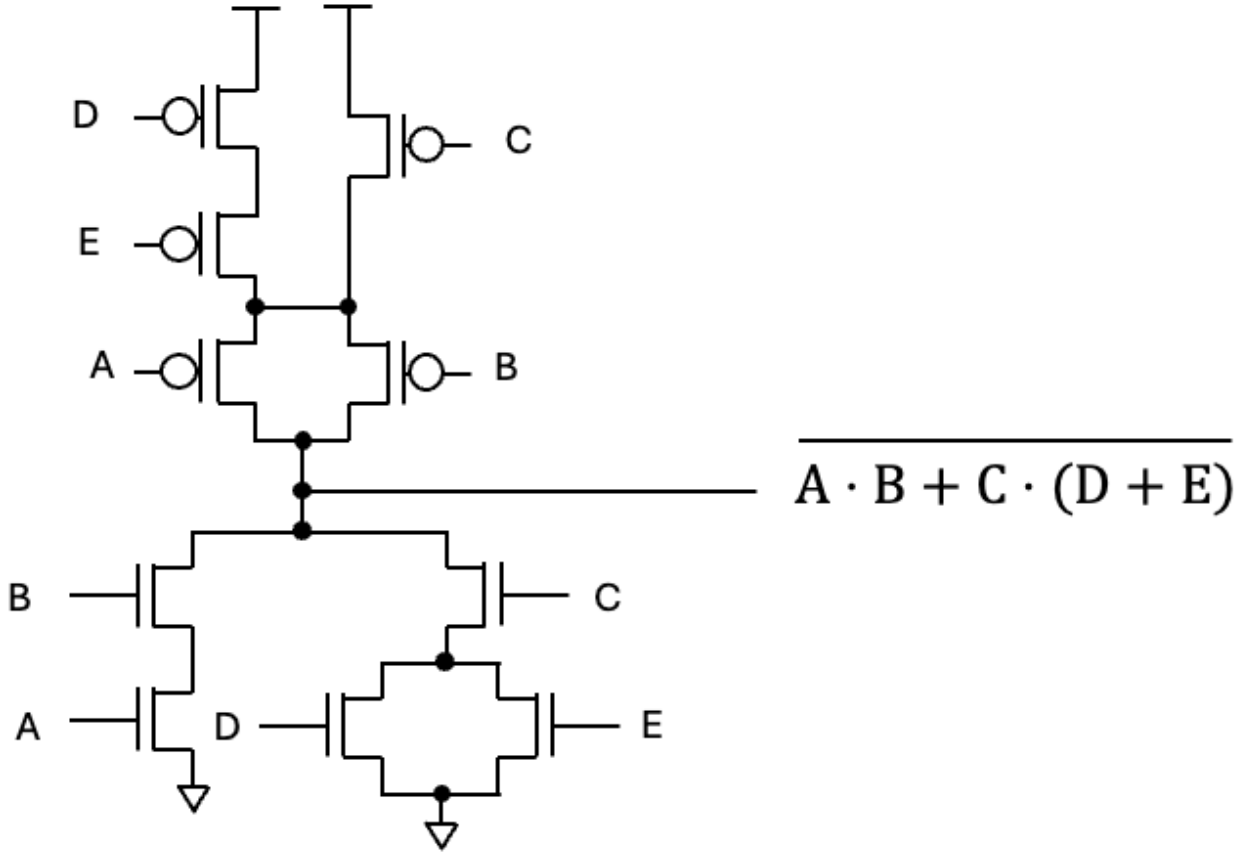






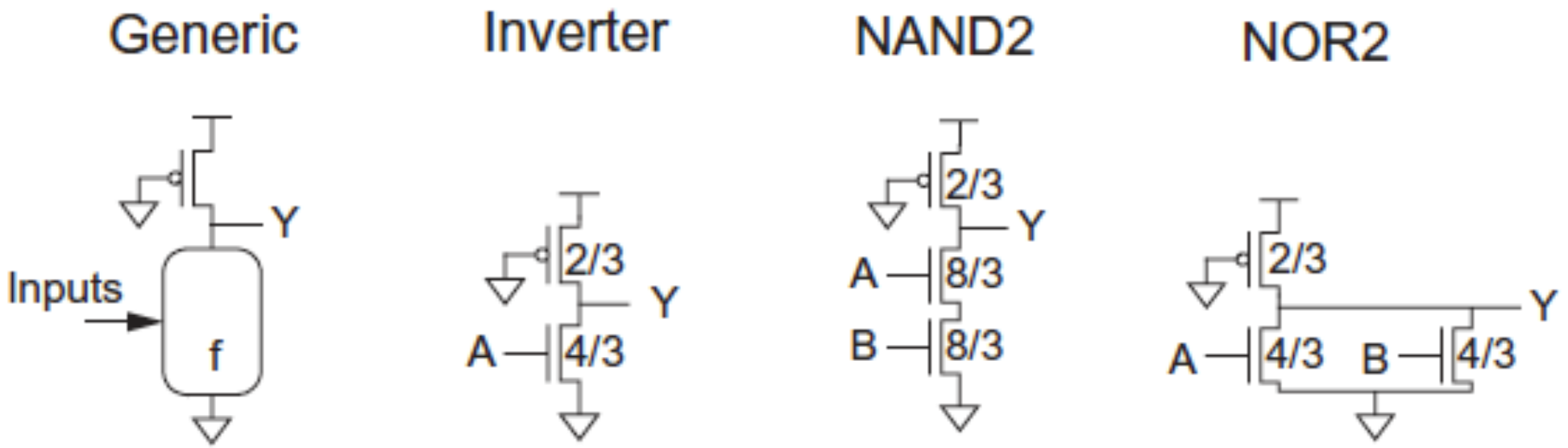
Complex Compound Gate





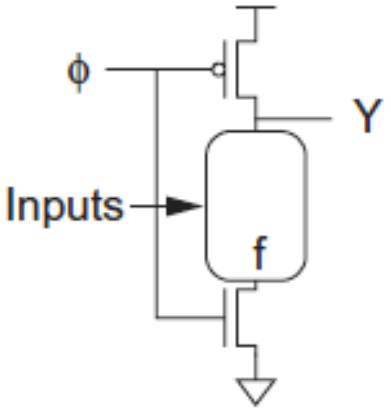
$$\overline{A \cdot B + C \cdot (D + E)}$$

Pseudo-nMOS Logic Gates

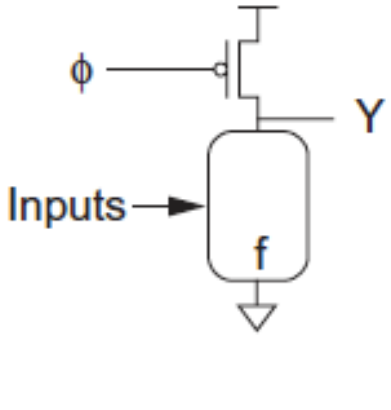


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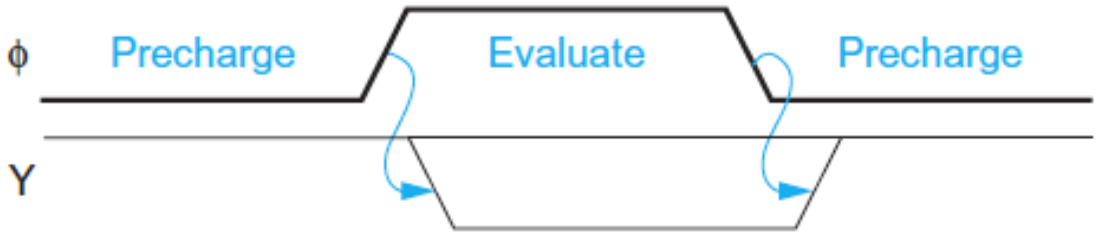
Dynamic Logic Gates

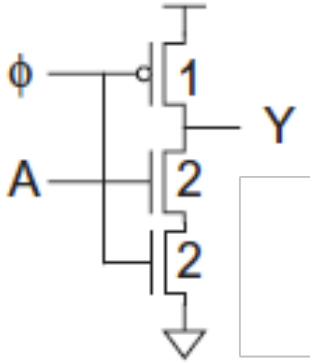


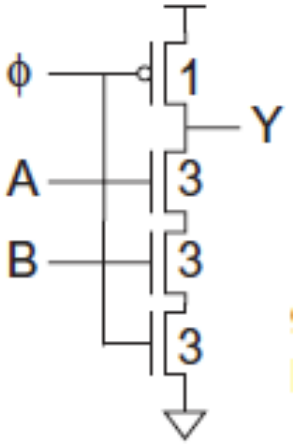
Footed

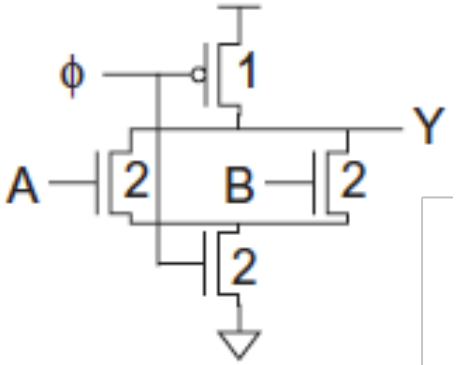


Unfooted



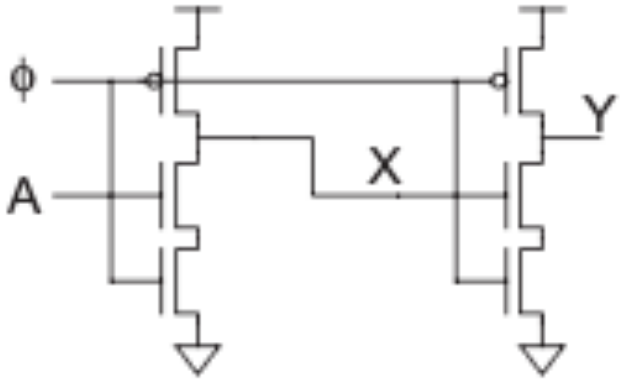


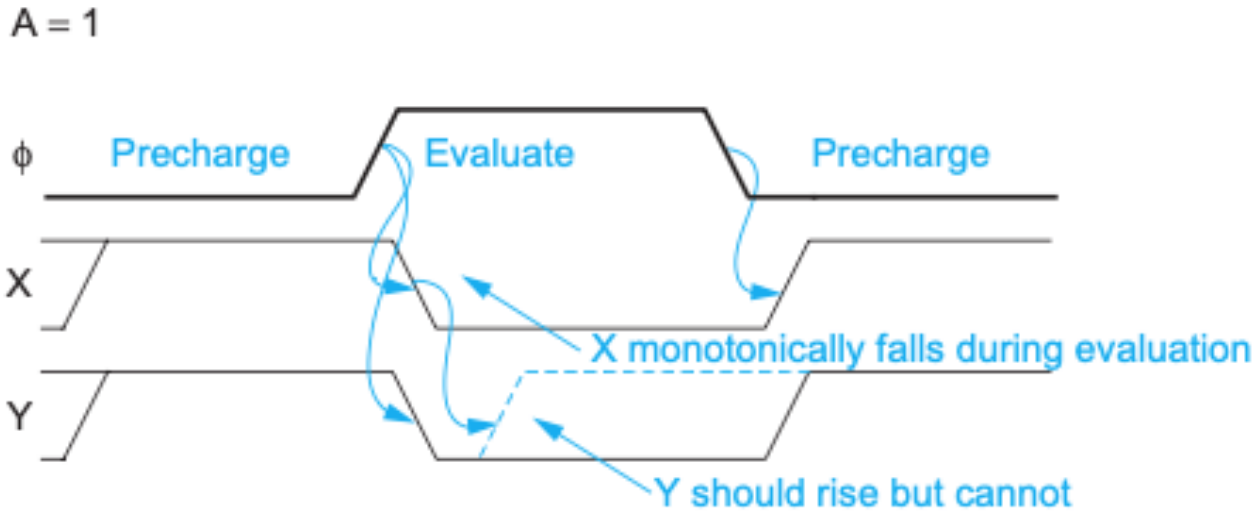




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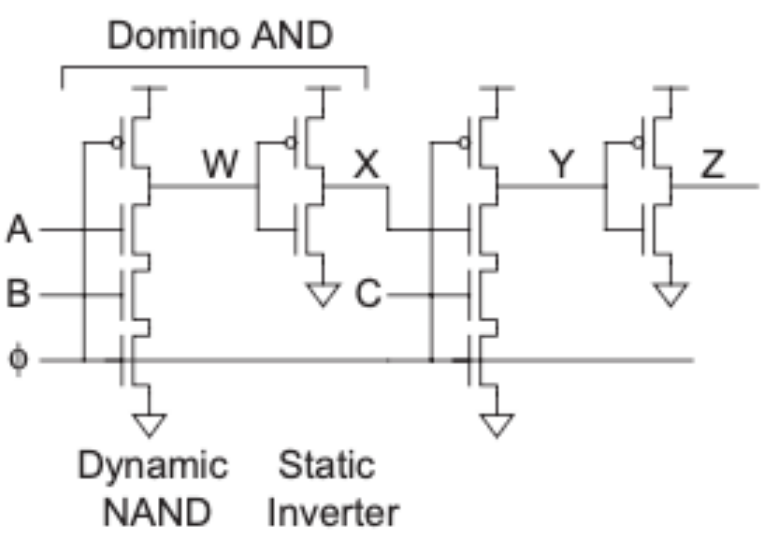
Problem with Dynamic Logic Gates

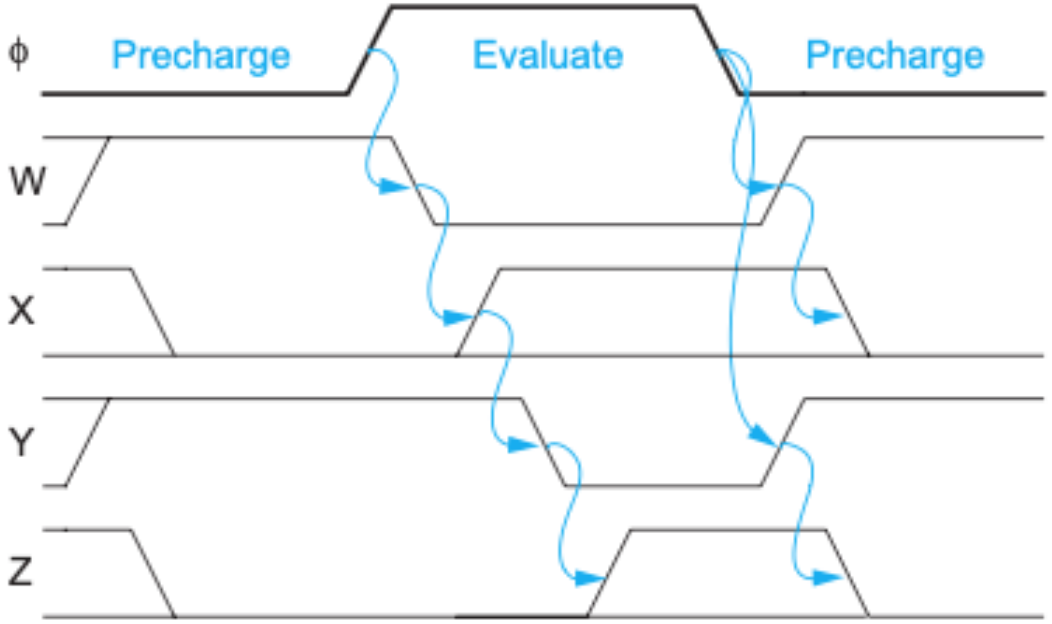


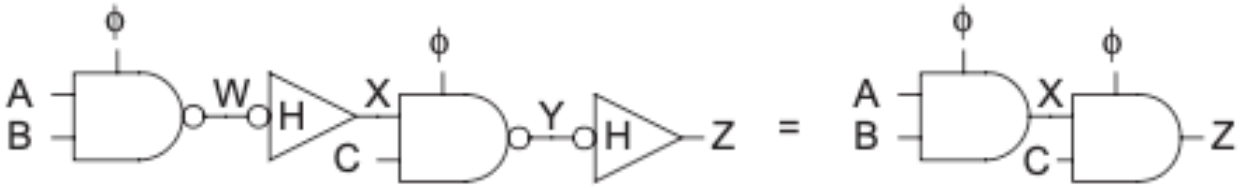


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Domino Logic

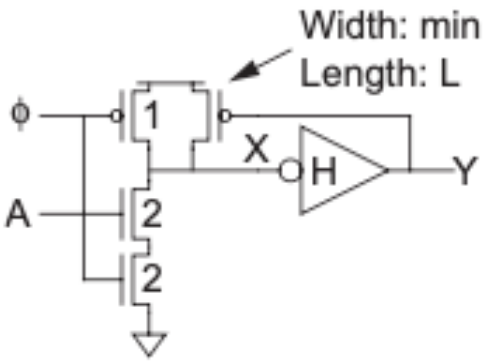


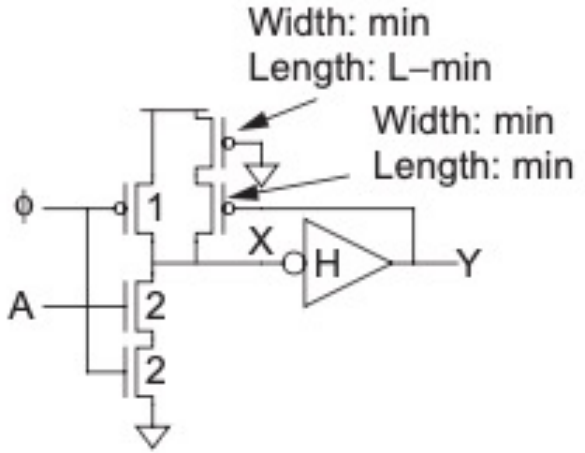


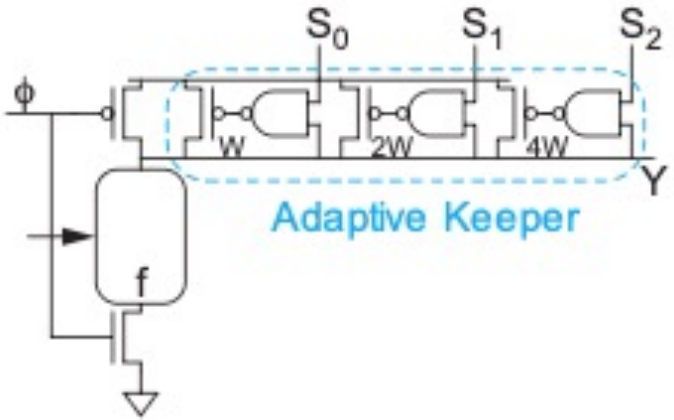


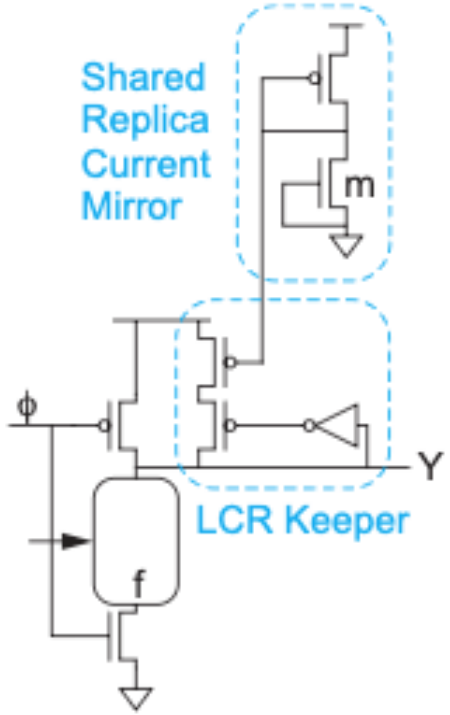
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Keepers for Dynamic Logic



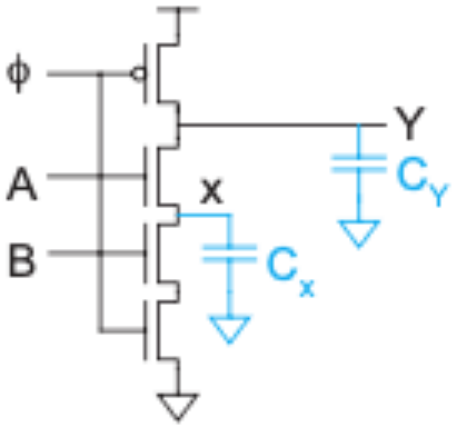


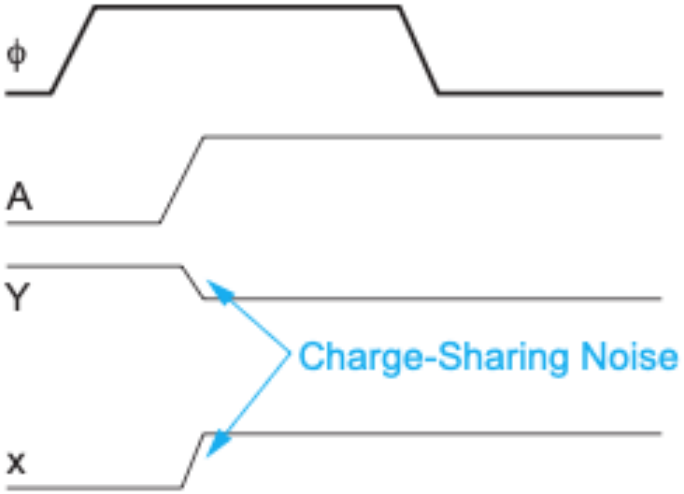


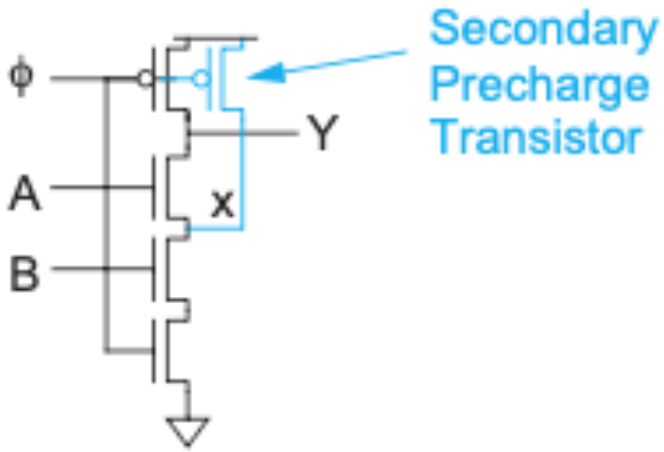


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Charge Sharing in Dynamic logic







W&H p345